

6. MASS AND BALANCE

6.1 WASHER RESERVOIR LOADING DATA

Reservoir Max Quantity	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Cleaning Fluid Mixture - 6 Liters	5.68	2800	15900

6.2 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Windshield Wiper/Washer System	105-80711	10.59	1870 (1750)	19803 (18533)
NOTE Above values in parentheses apply to Model S Variants.				

7. SYSTEM DESCRIPTION

Dual windshield wipers with washer capability are installed. The wiper portion includes a motor assembly driving two gearboxes, and two wiper arms with blades. The wash portion comprises a fluid reservoir located on the cargo compartment floor, an electric pump, a check valve, one sprayer assembly per wiper blade and the interconnecting hoses.

Operation of the system is controlled by use of a WIPER/OFF/WASH toggle switch on the center console switch panel. Placing the three-position WIPER/OFF/WASH switch in the WIPER position supplies 28V DC from the Auxiliary Bus via circuit breaker WIPER located on the switch panel, through a controller to the wiper motor. Placing the WIPER/OFF/WASH switch to the OFF position will automatically bring the wiper blades to their vertical park position.

The WIPER/OFF/WASH switch is spring-loaded to OFF when using the WASH position. Momentarily placing this switch in the WASH position first supplies 28V DC from the Auxiliary Bus via circuit breaker WASH located on the switch panel, through the controller to the washer pump motor and then after a short delay supplies power to the wiper motor. The pump motor is protected by a timer relay that automatically cuts off power to the pump after approx 2 seconds. Releasing the switch will then return the blades to their park position and reset the pump timer relay.